$\mathbb{R}^{2N_c+1}$, the W_i are independent Wiener processes, and D is the "deterministic part:"

$$D = \begin{bmatrix} \xi \boldsymbol{p} \\ -\boldsymbol{p} \\ -\boldsymbol{p} \cdot \nabla U - \xi \end{bmatrix}.$$

Then we calculate directly

$$D' = [D, Y_{N_c+1}] = \begin{bmatrix} \boldsymbol{p} \\ 0 \\ 1 \end{bmatrix},$$

and from this we see that $[D', Y_i] = \boldsymbol{e}_i$ for $i = 1, 2, \ldots, N_c$. Thus all the basis vectors of $\mathbb{R}^{2N_c+1}$ are found in the Lie algebra of the components of the Hörmander decomposition; as we know, this is the key step in the proof of ergodicity. Thus the method (8.33)–(8.35) is an alternative for molecular sampling dynamics, even though it is not based on a Newtonian model. Leaving the stochastic term off the final equation (i.e. in ξ) is easily shown to destroy the ergodicity property (the Hörmander condition is then violated).

Now consider the splitting into deterministic and OU parts. The deterministic part

$$\dot{\boldsymbol{q}} = \xi M^{-1}\boldsymbol{p}, \quad \dot{\boldsymbol{p}} = 0, \quad \dot{\xi} = -(\boldsymbol{M}^{-1}\boldsymbol{p}) \cdot \nabla U(\boldsymbol{q})/\mu, \tag{8.36}$$

preserves volume and the function $U(\boldsymbol{q}) + \mu\xi^2/2$ and hence also the distribution with density $\tilde{\rho}_A(\boldsymbol{q}, \xi) = e^{-\beta(U(\boldsymbol{q})+\mu\xi^2/2)}$. By the same token the OU processes collectively preserve $\tilde{\rho}_B(\boldsymbol{p}, \xi) = e^{-\beta(\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}/2+\mu\xi^2/2)}$. In combination the full dynamics preserves the extended canonical measure. If we iterate the two parts (assuming the deterministic part is solved exactly) the distribution will converge to the exact canonical distribution, due to especially to incorporation of $N_c + 1$ noise processes.

The OU process defining the fluctuations in $\boldsymbol{p}$ is completely decoupled from $\boldsymbol{q}$ and ξ, so it is perhaps most natural to think of the iteration as a random walk on the reduced space $(\boldsymbol{q}, \xi)$ where the directions are generated by an auxiliary OU process. Then the algorithm proceeds as follows: (1) choose a random direction $\boldsymbol{p}$ using the OU process, and (2) sample the reduced (stochastic) dynamics for $(\boldsymbol{q}, \xi)$ in the direction $\boldsymbol{p}$. We refer to this procedure as *stochastic line sampling* (SLS), for the following reason. With $\boldsymbol{p}$ fixed in step (2), the motion of $\boldsymbol{q}$ is confined to the straight line: $\boldsymbol{q} = \boldsymbol{q}_0 + \eta\boldsymbol{M}^{-1}\boldsymbol{p}$, where η is a scalar function of time. Inserting this coordinate change in (8.36) we see that the equations reduce to $\dot{\eta} = \xi$, $\dot{\xi} = -\partial/\partial\eta U(\boldsymbol{q}_0 + \eta\boldsymbol{M}^{-1}\boldsymbol{p})$. This is a one-degree of freedom system and hence is theoretically exactly solvable in quadratures, although the complexity of the force will render this impractical in calculations, and, as mentioned above, an additional stochastic process must be incorporated to drive ξ.